Simplicial Homotopy Theory

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Abstract

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1 Simplicial Homotopy Theory

Algebraic topology is a new subject which received a name change from combinatorial topology. At that time, combinatorics was highly involved in deriving topological invariants because the combinatorial structure of any invariants or spaces of study makes computation easy. In particular, a great deal of work has been put into the homotopy theory of simplicial spaces.

Nowadays, the study of combinatorial objects in topology is less prominent, but the category of simplicial sets still play a distinguished role in algebraic topology. Indeed the category of simplicial sets is the prototypical example of a model category (we have not seen it), as well as exhibiting a Quillen equivalence between the model of a simplicial set and some model category structure of **Top**. We will not explore the concept of model categories but will instead display foundational knowledge of it in order to develop the category of simplicial sets as a workable category when studying Model Categories. This means that we will develop the notion of homotopy, fibrations and cofibrations in this category.

1.1 Homotopy of Simplicial Sets

Definition 1.1.1 (Homotopy between Two Morphisms) Let X, Y be simplicial sets. Let $f, g : X \to Y$ be two morphisms. A simplicial homotopy from f to g is a morphism $\eta : X \times \Delta^1 \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} X \simeq X \times \Delta[0] \arrow[r, "\mathrm{id}_X \times \delta_1"] \arrow[dr, "f"'] & X \times \Delta[1] \arrow[d, dashed, "\exists \eta"] & X \simeq X \times \Delta[0] \arrow[l, "\mathrm{id}_X \times \delta_1"'] \arrow[dl, "g"] \\ & Y & \end{tikzcd}$$

We say that f and g are homotopic either if there is a homotopy from f to g, or there is a homotopy from g to f.

Definition 1.1.2 (Homotopy Relative to Subsets) Let X, Y be simplicial sets. Let $K \subseteq X$ be a simplicial subset. Let $\iota : K \hookrightarrow X$ denote the inclusion. Let $f, g : X \to Y$ be two morphisms. Let $\eta : X \times \Delta^1 \to Y$ be a homotopy from f to g. We say that ι is a homotopy relative to K if the following diagram commutes:

$$\begin{tikzcd} K \times \Delta^1 \arrow[r, "\mathrm{proj}_K"] \arrow[d, hook, "\iota \times \mathrm{id}_{\Delta^1}"'] & K \arrow[d, "\alpha"] \\ X \times \Delta^1 \arrow[r, "\eta"'] & Y \end{tikzcd}$$

where $\alpha = f|_K = g|_K$.

Proposition 1.1.3 Let X, Y be simplicial sets. Let $f, g : X \to Y$ be morphisms. Then there exists a homotopy from g to f if and only if there exists a family of morphisms $h_i^n : X_n \to Y_{n+1}$ such that the following are true.

- $d_0^n \circ h_0 = f_n$
- $d_0^{n+1} \circ h_1 = g_n$
- The composition

$$d_i \circ h_j = \begin{cases} h_{j-1} \circ d_i^{n+1} & \text{if } 0 \leq i < j \leq n \\ d_i \circ h_{i-1} & \text{if } i = j \neq 0 \\ h_j \circ d_{i-1} & \text{if } i > j + 1 \end{cases}$$

- The composition

$$s_i \circ h_j = \begin{cases} h_{j+1} \circ s_i & \text{if } i \leq j \\ h_j \circ s_{i-1} & \text{if } i > j \end{cases}$$

The homotopy relation is in general not an equivalence relation. However, when Y is a Kan complex, this is true.

Proposition 1.1.4 Let X be a simplcial set. Let Y be a Kan complex. Then the relation of homotopic morphisms from X to Y is an equivalence relation.

Definition 1.1.5 (Homotopy Inverses) Let X, Y be simplicial sets. Let $f : X \to Y$ be a morphism. A homotopy inverse of f is a morphism $g : Y \to X$ such that there exist a homotopy from $g \circ f$ to id_X and a homotopy from $f \circ g$ to id_Y.

Definition 1.1.6 (Homotopy Equivalences) Let X, Y be simplicial sets. Let $f : X \to Y$ be a morphism. We say that f is a homotopy equivalence between X and Y if f admits a homotopy inverse.

Lemma 1.1.7 Let X, Y be simplicial sets. Let $f : X \to Y$ be a homotopy equivalence. Then the homotopy inverse of f is unique up to homotopy.

1.2 Homotopy between Vertices

We can also talk about homotopies between vertices of a simplicial set.

Definition 1.2.1 (Homotopy between Vertices) Let X be a simplicial set. Let v_0, v_1 be vertices of X. A homotopy from v_0 to v_1 is a homotopy from the inclusion $\Delta^0 \cong \{v_0\} \hookrightarrow X$ to the inclusion $\Delta^0 \cong \{v_1\} \hookrightarrow X$.

Lemma 1.2.2 Let X be a simplicial set. Let v_0, v_1 be vertices of X. Then there exists a homotoyp from v_0 to v_1 if and only there exists and edge of X from v_0 to v_1.

We can generalize this to n-simplicies.

1.3 The Simplicial Homotopy Groups

Because homotopies from X to Y is an equivalence relation only when Y is a Kan complex, we can only define simplicial homotopy groups for Kan complexes. This notion can easily be extended to arbitrary simplicial sets by replacing the simplicial set with a fibrant replacement (and hence becomes a Kan complex).

Definition 1.3.1 (Simplicial Homotopy Groups) Let X be a Kan complex. Let $v \in X$ be a vertex of X. Define the simplicial homotopy groups of (X, v) as follows:

- For $n \geq 1$, define the nth simplicial homotopy group $\pi_n^{\text{sSet}}(X, v)$ of X at v to be the set of homotopy classes of maps $[\alpha : \Delta^n \to X]$ relative to boundary $\partial\Delta^n$. In other words, the following diagram commutes:

$$\begin{tikzcd}
\partial\Delta^n \arrow[r, "!"] \arrow[d, hook] & \Delta^0 \arrow[d, "v"] \\
\Delta^n \arrow[r, "\alpha"'] & X
\end{tikzcd}$$

- Define the 0th simplicial homotopy group $\pi_0^{\text{sSet}}(X)$ of X to be the set of homotopy classes of vertices of X.

If X is a general simplicial set, define the simplicial homotopy group of X to be

$$\pi_n^{\text{sSet}}(X, v) = \pi_n^{\text{sSet}}(PX, Pv)$$

where PX is a fibrant replacement of X.

Theorem 1.3.2 Let X be a Kan complex. Let $f, g : \Delta^n \to X$ be two representatives of two elements in $\pi_n(X, v)$ for $n \geq 1$. Then the following data

$$v_i = \begin{cases} s_0^n(v) & \text{if } 0 \leq i \leq n-2 \\ f & \text{if } i = n-1 \\ g & \text{if } i = n+1 \end{cases}$$

defines a horn $\Lambda_i^{n+1} \to X$. Such a map extends to a map $\theta : \Delta^{n+1} \to X$. Define

$$[f] \cdot [g] = d_n \theta$$

Then such an operation is well defined on the equivalence class. Moreover, it defines a group operation on $\pi_n^{\text{sSet}}(X, v)$.

Theorem 1.3.3 Let X be a Kan complex. Then for $n \geq 2$, the above group structure on $\pi_n^{\text{sSet}}(X, v)$ is abelian.

Theorem 1.3.4 Let X be a simplicial set. Let $v \in X_0$. Then there is an isomorphism

$$\pi_n^{\text{sSet}}(X, v) \cong \pi_n(|X|, v)$$

for all $n \in \mathbb{N}$.

2 Homotopy Limits and Colimits

2.1 The Standard Homotopy Pullback and Pushout

Definition 2.1.1 (The Standard Homotopy Pullbacks) Let the following be a diagram of simplicial sets and morphisms.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

Define the standard homotopy pullback to be the simplicial set

$$\text{Holim}_Q(X \to Z \leftarrow Y) = X \times_Z^h Y = X \times_{\text{Hom}_{\mathbf{sSet}}(\{0\}, Z)} \text{Hom}_{\mathbf{sSet}}(\Delta^1, Z) \times_{\text{Hom}_{\mathbf{sSet}}(\{1\}, Z)} Y$$

Here we identify the vertices of Δ^1 as 0 and 1, and we use the isomorphism $\mathcal{C} \cong \text{Hom}_{\mathbf{sSet}}(\Delta^0, Z)$.

Kerodon 3.4.0.3

Lemma 2.1.2 Let the following be a diagram of simplicial sets and morphisms, where Z is a Kan complex.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

Then the following diagram commutes up to homotopy:

$$\begin{tikzcd}
\text{Holim}_Q(X \to Z \leftarrow Y) \arrow[r, "\text{proj}"] \arrow[d, "\text{proj}"'] & X \arrow[d, "f"] \\
Y \arrow[r, "g"'] & Z
\end{tikzcd}$$

2.2 Homotopy Pullback and Pushout Squares

2.3 Recognizing Homotopy Pullbacks and Pushouts

Proposition 2.3.1 (Kerodon 3.4.0.7) Let the following be a diagram in simplicial sets.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

If Z is a Kan complex and one of f or g is a Kan fibration, then the following are true.

- The following diagram
$$\begin{tikzcd}
X \times_Z Y \arrow[r, "\text{pr}"] \arrow[d, "\text{pr}"'] & X \arrow[d, "f"] \\
Y \arrow[r, "g"'] & Z
\end{tikzcd}$$
is a homotopy pullback square.
- The inclusion map
$$X \times_Z Y \hookrightarrow X \times_Z^h Y$$
is a weak homotopy equivalence.
- $X \times_Z Y$ is a homotopy pullback of the diagram.

Secret: if the diagram is fibrant in the diagram category, then the homotopy pullback is the same as the standard pullback up to weak equivalence.

3 Fibrations and Cofibrations

3.1 Lifting Properties

Definition 3.1.1 (The Left Lifting Property) Let $A, B, X, Y \in \mathbf{sSet}$ be simplicial sets. Let $p : X \to Y$ and $i : A \to B$ be morphisms. We say that i has the has the left lifting property with respect to p if the following is true. For all morphisms $f : A \to X$ and $g : B \to Y$ such that $p \circ f = g \circ i$, there exists $h : B \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r, "g"'] \arrow[ur, dashed, "\exists h"] & Y \end{tikzcd}$$

Definition 3.1.2 (The Right Lifting Property) Let $A, B, X, Y \in \mathbf{sSet}$ be simplicial sets. Let $p : X \to Y$ and $i : A \to B$ be morphisms. We say that p has the has the right lifting property with respect to i if the following is true. For all morphisms $f : A \to X$ and $g : B \to Y$ such that $p \circ f = g \circ i$, there exists $h : B \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r, "g"'] \arrow[ur, dashed, "\exists h"] & Y \end{tikzcd}$$

3.2 Kan Fibrations

Definition 3.2.1 (Kan Fibrations) Let $f : X \to Y$ be a morphism of simplicial sets. We say that f is a Kan fibration if the following condition is satisfied: For every commutative diagram:

$$\begin{tikzcd} \Lambda^n_k \arrow[r] \arrow[d, hook] & X \arrow[d, "f"] \\ \Delta^n \arrow[r] & Y \end{tikzcd}$$

where $n \geq 1$ and $0 \leq k \leq n$, there exists a lift $\Delta^n \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} \Lambda^n_k \arrow[r] \arrow[d, hook] & X \arrow[d, "f"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

Lemma 3.2.2 Let X be a simplicial set. Then X is a Kan complex if and only if the unique map $X \to *$ is a Kan fibration.

Closed under retracts, pullbacks, filtered colimits, products with standard simplicies, composition

3.3 Anodyne Morphisms

They are the acyclic cofibrations of sSet with standard model structure.

Definition 3.3.1 (Anodyne Morphisms) Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a morphism. We say that f is anodyne if f has the left lifting property with respect to all Kan fibrations:

$$\begin{tikzcd} A \arrow[r] \arrow[d, "\text{Kan Fib.}"'] & X \arrow[d, "f"] \\ B \arrow[r] \arrow[ur, dashed, "\exists h"] & Y \end{tikzcd}$$

3.4 Trivial Kan Fibrations

They are the acyclic fibrations of sSet with standard model structure.

Definition 3.4.1 (Trivial Kan Fibrations) Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a morphism. We say that f is a trivial Kan fibration if f has the right lifting property with respect to all inclusions $\partial\Delta^n \hookrightarrow \Delta^n$:

$$\begin{tikzcd} \partial\Delta^n \arrow[r] \arrow[d, hook] & X \arrow[d, "f"] \\ \Delta^n \arrow[r] \arrow[ur, dashed, "\exists k"] & Y \end{tikzcd}$$

Proposition 3.4.2 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a morphism. Then f is a trivial Kan fibration if and only if f has the right lifting property with respect to all monomorphisms in $\mathbf{sSet}$.

Corollary 3.4.3 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a morphism. If f is a trivial Kan fibration, then f admits a section. This means that there exists a morphism $s : Y \to X$ such that

$$f \circ s = \mathrm{id}_Y$$

Moreover, $s \circ f : X \to X$ is homotopic to the identity map id_X.

Proposition 3.4.4 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a trivial Kan fibration. If X is a Kan complex, then Y is a Kan complex.

Proposition 3.4.5 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a morphism. If f is a trivial Kan fibration, then the induced morphism

$$f \circ - : \mathrm{Hom}_{\mathbf{sSet}}(A, X) \to \mathrm{Hom}_{\mathbf{sSet}}(A, Y)$$

is a trivial Kan fibration for all $A \in \mathbf{sSet}$.

3.5 Trivial Kan Fibrations and Contractible Kan Complexes

Proposition 3.5.1 Let X be a simplicial set. Then the following are equivalent.

- X is a contractible Kan complex (every morphism $\partial\Delta^n \to X$ has an extension $\Delta^n \to X$)
- The unique map $X \to \Delta^0$ is a trivial Kan fibration.
- For every monomorphism $i : A \to B$ of simplicial sets and every morphism of simplicial sets $f : A \to X$, there exists a map $g : B \to X$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, hook] & X \\ B \arrow[ur, dashed, "g"'] & \end{tikzcd}$$

Proposition 3.5.2 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \to Y$ be a trivial Kan fibration. If X is a contractible Kan complex, then Y is a contractible Kan complex.

4 Topics in Kan Complexes

4.1 Ex^∞ and the Subdivision Functor

4.2 Weak Factorization

Proposition 4.2.1 Let X, Y be simplicial sets. Let $f : X \to Y$ be a morphism. Then there exists a simplicial set $Q(f)$ and maps $f' : X \to Q(f)$ and $f'' : Q(f) \to Y$ such that f' is anodyne, f'' is a Kan fibration and the following diagram commutes:

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, dashed, "\exists f'"'] & & Y \\ & Q(f) \arrow[ur, dashed, "\exists f''"'] & \end{tikzcd}$$

Proposition 4.2.2 Let X, Y be simplicial sets. Let $f : X \to Y$ be a morphism. Then there exists a choice of $Q(f)$ such that Q defines a functor

$$Q : \mathrm{Hom}_{\mathbf{Cat}}(\Delta^1, \mathbf{sSet}) \to \mathbf{sSet}$$

that commutes with filtered colimits.

4.3 (Co)Fibrant Replacements

Definition 4.3.1 (Fibrant Replacements) Let X be a simplicial set. A fibrant replacement of X is a Kan complex QX such that the unique map $X \to *$ factorizes into $X \to QX \to *$ where $X \to QX$ is anodyne.

Secret: these are just the fibrant replacements in model category.

Lemma 4.3.2 Let X be a simplicial set. Then the following are all fibrant replacements of X.
- $Q_X = Q(X \to *)$
- $\mathrm{Ex}^\infty(X)$